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8 August 1983
SUPERSEDING
MIL-A-46108A
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MILITARY SPECIFICATION

ARMOR: TRANSPARENT, GLASS; GLASS/PLASTIC; PLASTIC LAMINATES (GENERAL SPECIFICATION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for laminated, transparent armor used in armored military vehicles and aircraft.

1.2 Classification. Transparent armor is a technical term for protective transparencies commonly called "ballistic resistant windows." Transparent armor constructions shall be classified as follows:

Type I - Glass/plastic laminated construction - consists of one or more plies of glass as an exterior surface bonded to a plastic backing material.

Type II - Glass-clad plastic "sandwich" construction - consists of two or more glass outer layers bonded to a plastic inner core.

Type III - All plastic laminated construction.

Type IV - All glass laminated construction.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Materials and Mechanics Research Center, ATTN: DRXMR-SMS, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

DD-G-451	Glass, Float or Plate, Sheet, Figured (Flat, for Glazing, Mirrors and other Uses)

MILITARY

MIL-G-3787	Glass, Laminated, Flat (Except Aircraft)
MIL-P-5425	Plastic, Sheet, Acrylic, Heat Resistant
MIL-G-5485	Glass; Laminated, Flat, Bullet-Resistant
MIL-P-8184	Plastic Sheet, Acrylic, Modified
MIL-B-11352	Block, Vision: Bullet-Resistant
MIL-P-25374	Plastic Sheet, Acrylic, Modified, Laminated
MIL-G-25667	Glass, Monolithic, Aircraft Glazing
MIL-P-25690	Plastic, Sheets and Parts, Modified Acrylic Base, Monolithic, Crack Propagation Resistant
MIL-G-25871	Glass, Laminated, Aircraft Glazing
MIL-P-46144	Plastic Sheet, Polycarbonate
MIL-P-83310	Plastic Sheet, Polycarbonate, Transparent

STANDARDS

MILITARY

MIL-STD-444	Nomenclature and Definitions in the Ammunition Area
MIL-STD-810	Environmental Test Methods
MIL STD-1288	Aircrew Protection Requirements, Nonnuclear Weapons Threat

HANDBOOKS

MILITARY

MIL-HDBK-17 (Part II)	Plastics for Aerospace Vehicles
MIL-HDBK-336	Survivability, Aircraft, Nonnuclear, General Criteria - Volume I
MIL-HDBK-722	Glass

(Copies of specifications, standards and handbooks required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1003	Haze and Luminous Transmittance of Transparent Plastics
ASTM D 1044	Resistance of Transparent Plastics to Surface Abrasion
ASTM D 3951	Commercial Packaging
ASTM D 4093	Photoelastic Measurements of Birefringence and Residual Strains in Transparent or Translucent Plastic Materials
ASTM F 428	Intensity of Scratches on Aerospace Glass Enclosures
ASTM F 520	Environmental Resistance of Aerospace Transparencies
ASTM F 521	Transparent Laminates For Bond Integrity
ASTM F 548	Intensity of Scratches on Aerospace Transparent Plastics
ASTM F 735	Abrasion Resistance of Transparent Plastics and Coatings Using the Oscillating Sand Method

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction. Unless otherwise specified (see 6.2), a preproduction sample shall be required. The preproduction sample shall conform to all requirements in this specification.

3.2 Materials. The transparent armor shall consist of plies of glass and/or plastic and adhesive interlayer materials as specified in the contract or order (see 1.2, 6.2 and 6.4).

3.2.1 Glass. Glass used in transparent armor constructions shall be as specified in the contract or order (see 6.2). The following documents may be used to specify glass requirements:

a. DD-G-451
b. MIL-G-25667
c. MIL-HDBK-722

3.2.2 Plastic. Plastic used in transparent armor constructions shall be either polycarbonate or acrylic sheet as specified in the contract or order (see 6.2). The transparent plastic sheet shall be tested for abrasion resistance in accordance with ASTM D 1044, ASTM F 735 or other method as specified in the contract or order (see 6.2).

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3.2.2.1 Polycarbonate plastic. Unless otherwise specified (see 6.2), the polycarbonate plastic sheet used in construction of transparent armor fabricated under this specification shall be ultraviolet (UV) stabilized, shall conform to MIL-P-46144, and shall have a mar-resistant coating applied to its exposed surface. Polycarbonate plastic sheet intended for use in U.S. Air Force windshields and canopies shall conform to MIL-P-83310 (USAF).

3.2.2.2 Acrylic plastic. Unless otherwise specified (see 6.2), the acrylic plastic sheet used in construction of transparent armor fabricated under this specification shall be as-cast or stretched. The following documents may be used to specify requirements for acrylic plastic sheet:

a. MIL-P-5425
b. MIL-P-8184
c. MIL-P-25690

3.2.3 Adhesive interlayer materials. The adhesive interlayer materials for bonding glass-to-glass, glass-to-plastic or plastic-to-plastic shall be chemically compatible with the surfaces which are being bonded. When specified (see 6.2), the bond integrity of the transparent laminates shall be determined (see 4.5). The interlayer materials used in the transparent armor shall be as specified in the contract or order (see 6.2). Interlayer materials may be polyvinyl butyral, cast-in-place urethane, proprietary materials, sheet form urethane and other materials (see 6.4). Polyvinyl butyral shall not be used in contact with polycarbonate because its plasticizer may craze polycarbonate.

3.3 Design.

3.3.1 Size and shape. The size and shape of the transparent armor shall be as specified in the contract or order (see 6.2). Transparent armor may be flat or curved and special sizes and shapes may also be designed to meet specific needs.

3.3.2 Weight and thickness. The armor laminates shall be of the minimum weight and thickness consistent with the ballistic protection requirements specified in the contract or order (see 6.2), and shall not exceed values specified in the contract or order.

3.3.3 Construction. The construction of the transparent armor shall be as specified in the contract or order (see 6.2). Glass laminates may be constructed in accordance with MIL-G-3787, MIL-G-5485 or MIL-G-25871. Plastic laminates may be constructed in accordance with MIL-P-25374.

3.4 Ballistic limit. The ballistic limit of transparent armor used in military weapon systems shall be as specified in the contracts or detailed specifications for the particular weapon systems. Ballistic requirements for transparent armor shall be stated in accordance with the terminology of MIL-HDBK-336-1, Section 6 (Threat System Description) and MIL-STD-444 for standardization to the maximum extent practicable.

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3.5 Optical.

3.5.1 Luminous transmittance and haze. Requirements for luminous transmittance and haze shall be as specified in the contract or order (see 6.2).

3.5.2 Optical deviation. Requirements for optical deviation shall be as specified in the contract or order (see 6.2). Optical deviation shall be defined and measured as specified in the contract or order (see 6.2).

3.5.3 Optical distortion. Requirements for optical distortion shall be as specified in the contract or order (see 6.2). Optical distortion shall be defined and measured as specified in the contract or order (see 6.2).

3.5.4 Birefringence. Requirements for birefringence shall be as specified in the contract or order (see 6.2). When specified (see 6.2), photoelastic measurements of birefringence and residual strains shall be determined for the transparent plastic materials used in the armor constructions (see 4.5).

3.6 Environmental resistance. The transparent armor shall not crack, cloud, delaminate, or show any other evidence of deterioration when tested for environmental resistance (see 4.5). Additional requirements for the environmental resistance of the transparent armor shall be as specified in the engineering drawing, detailed specification or contract in accordance with MIL-STD-810 for the following environmental conditions:

a. Low pressure (altitude)
b. High temperature
c. Low temperature
d. Temperature shock
e. Temperature - altitude
f. Solar radiation (sunshine)
g. Rain
h. Humidity
i. Fungus
j. Salt fog
k. Dust (fine sand)
l. Explosive atmosphere
m. Chemical resistance
n. Acceleration
o. Vibration
p. Acoustical noise
q. Shock
r. Temperature - humidity - altitude
s. Gunfire vibration, aircraft

3.7 Allowable defects. For inspection purposes, the laminate is divided into three zones or grading areas. Area "A" is the edge to be concealed at assembly and is specified by the procuring activity (see 6.2); "B" is the area within 2 inches (50 mm) of the concealed edges; and "C" is the remaining central area. Maximum allowable defects shall conform to table I.

3.8 Workmanship. Unless otherwise specified (see 6.2), minor imperfections that do not affect serviceability shall be permitted.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor or may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. Unless otherwise specified (see 6.2), the preproduction sample shall be manufactured by the same methods proposed for the manufacture of subsequent production lots. The preproduction sample shall be subjected to all examinations and tests specified herein. If the preproduction sample is rejected, it will be returned to the contractor. Before submitting another preproduction sample, the contractor shall furnish the Government full particulars concerning previous rejection and the action taken to correct the defects. Subsequent quantities shall not be considered for acceptance until the approval of the preproduction sample has been obtained.

4.3 Classification of tests. All tests of transparent armor under this specification shall be classified as preproduction tests (see 4.5).

4.4 Examination.

4.4.1 Visual. The transparent armor shall be visually examined for compliance with the requirements for workmanship (see 3.8).

4.4.2 Packaging. Prior to shipment, examination shall be made to determine compliance with the requirements of section 5.

4.5 Test methods (except ballistics). When specified (see 6.2), transparent armor laminates shall be tested in accordance with the test methods in table II or other test methods approved by the procuring activity.

4.6 Ballistic test.

4.6.1 Marking of test samples. Transparent armor samples submitted for ballistic testing shall be clearly marked "IMPACT SIDE" on the side to be impacted by the test projectile.

4.6.2 Test range. Ballistic qualification or acceptance tests shall be performed at a ballistic facility selected by the procuring activity.

4.6.3 Procedure. The ballistic test procedure shall be as specified by the procuring activity (see 6.2)

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with the applicable packaging standard or packaging data sheet, specified by the procuring activity (see 6.2).

5.2 Industrial packaging. When specified (see 6.2), packaging shall be in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The transparent armor covered by this specification is intended for use in military aircraft, ships, tactical vehicles, or other applications for ballistic protection against small arms projectiles (see 6.3.1). Transparent armor constructions provide protection ranging from small caliber handguns to military rifles. Advanced glass/plastic vision blocks have been fabricated for military fighting vehicles (see MIL-B-11352). Transparent armor is also installed in limousines used by political leaders and high-ranking military officers. Transparent armor is used as crew station armor on military helicopters (see MIL-STD-1288).

6.2 Acquisition data. Acquisition documents should specify the following:

a. Title, number and date of specification
b. Destination of preproduction test sample if required (see 3.1)
c. Materials requirements (see 3.2)
d. Glass requirements (see 3.2.1)
e. Plastic requirements (see 3.2.2)
f. Interlayer materials (see 3.2.3)
g. Size and shape (see 3.3.1)
h. Weight and thickness (see 3.3.2)
i. Construction (see 3.3.3)
j. Ballistic limit (see 3.4)
k. Luminous transmittance and haze requirements (see 3.5.1)
l. Optical deviation requirements (see 3.5.2)
m. Optical distortion requirements (see 3.5.3)
n. Birefringence requirements (see 3.5.4)
o. Area to be concealed at assembly (see 3.7)
p. Requirements pertaining to occasional minor imperfections if other than in 3.7
q. Test methods to be conducted (see 4.5 and 4.6) and minimum acceptance values
r. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
s. Applicable drawings if required

6.3 Definitions.

6.3.1 Small arms. See MIL-STD-444; MIL-STD-2089, 5.1.1.1-3; and MIL-HDBK-336-1, 6.1.1.1.2.

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6.4 Interlayer materials. Various interlayer materials have been used to adhesively bond plies of transparent materials (glasses and plastics) in the fabrication of transparent armor. Military Handbook MIL-HDBK-17 Part II Chapter 5 describes some of the commonly used interlayer materials; however, as technology progresses, companies are developing proprietary materials that can be used as an interlayer.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 20

Review Activities:
Army - GL, AR, TE, SM, AV
Navy - MC
Air Force - 43, 99

User Activities:
Army - AT
Navy - SH

Preparing Activity:
Army - MR

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TABLE I. Maximum allowable defects.

Types of defects	Allowable defects 1/	
	Area "C"	Area "B"
Gaseous inclusions	1/16 in (1.6 mm)	3/32 in (2.4 mm)
Open gaseous inclusions	3/64 in (1.2 mm)	1/16 in (1.6 mm)
Stones and knots	1/32 in (0.8 mm)	3/64 in (1.2 mm)
Ream and string	Shall not be evident at an angle greater than 30° between line of sight and glass surface, with indirect daylight.	
Scratches and rubs	Use ASTM F 548 for plastics and ASTM F 428 for glass.	
Digs	1/16 in (1.6 mm)	3/32 in (2.4 mm)
Crush	Shall not be detectable at distances greater than 10 ft (3.0 m), with indirect daylight.	
Lint and hair	Shall not be detectable at distances greater than 3 ft (0.9 m) looking perpendicular to the glass, with indirect daylight.	
Inside dirt	3/32 in (2.4 mm)	3/32 in (2.4 mm)
Surface defects	Shall not be detectable at distances greater than 10 ft (3.0 m).	
Interlayer striae	Shall not be detectable at distances greater than 10 ft (3.0 m) looking perpendicular to the glass, with indirect daylight.	

1/ All discrete defects shall be separated by at least 12 in (30 cm). The defects listed are maximum permissible. Defects not specifically mentioned shall be compared to the defect they most closely resemble.

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TABLE II. Test methods[1/]

Test	Number
Abrasion resistance	ASTM D 1044 or ASTM F 735
Birefringence	ASTM D 4093
Bond integrity	ASTM F 521
Environmental resistance	ASTM F 520 and MIL-STD-810
Luminous transmittance and haze	ASTM D 1003

1/ These tests shall be conducted as specified by the procuring activity (see 6.2).

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b. Recommended Wording:

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